

IE-16: A counterexample and the failure of every universal subset constant

Sidney Holden

Center for Computational Biology, Flatiron Institute

Simons Foundation

12 September 2026

Abstract

The proposed $4/\pi$ subset bound is false. Let $\omega = e^{2\pi i/3}$ and $L = \{\omega^a + 10^{-3}\omega^b : 0 \leq a, b \leq 2\}$. At degree $k = 4$, this nine-point set satisfies

$$\frac{M_4(L)}{\max_{S \subset L, |S|=5} M_4(S)} > \frac{13}{10} > \frac{4}{\pi}.$$

The proof is analytic and covers every subset by its cluster occupancy. An additional exact-arithmetic certificate checks all 126 subsets and proves optimality of the full-set polynomial by positive weighted orthogonality. The actual ratio is approximately 1.33299891979475829424. A three-cluster amplification lemma gives a stronger conclusion: no finite constant, independent of dimension, degree and the points, can satisfy the proposed type of bound. For every positive integer r and every $\eta > 0$, examples with $n = 3^r$ and $k = (3^r - 1)/2$ have ratio greater than $(2/\sqrt{3})^r - \eta$.

1 The question and the conclusions

For a finite set $E \subset \mathbb{C} \setminus \{0\}$, put

$$M_k(E) = \min\{\|p\|_E : p \in \mathbb{C}[z], \deg p \leq k, p(0) = 1\}, \quad \|p\|_E = \max_{z \in E} |p(z)|.$$

For $|E| \geq k + 1$, define

$$B_k(E) = \max_{S \subseteq E, |S|=k+1} M_k(S), \quad R_k(E) = \frac{M_k(E)}{B_k(E)}.$$

All denominators used here are positive: a polynomial of degree at most k with value one at zero cannot vanish on $k + 1$ distinct nonzero points. The finite-dimensional minima exist because evaluation on at least $k + 1$ distinct points is injective.

IE-16 asks whether $R_k(E) \leq 4/\pi$ for every set of $n \geq 3$ distinct nonzero complex numbers and every $1 \leq k \leq n - 2$.¹ The original normal-GMRES paper formulates a universal-constant conjecture and proposes $4/\pi$ as the candidate.² The following two theorems give negative answers, respectively, to the stated constant and to the possibility of any universal finite replacement.

Theorem 1.1 (Explicit counterexample). *Let*

$$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \varepsilon = \frac{1}{1000}, \quad L_\varepsilon = \{\omega^a + \varepsilon\omega^b : a, b \in \{0, 1, 2\}\}.$$

¹Canonical statement: A. Townsend, *Open Problems in Numerical Linear Algebra*, entry IE-16, accessed 12 September 2026; Source 1 below.

²J. Liesen and P. Tichý, *The worst-case GMRES for normal matrices*, BIT 44 (2004), 79–98, §3.2.2, especially (3.16), pp. 91–92; Source 2 below.

Then L_ε consists of nine distinct nonzero points, $k = 4$ is admissible, and

$$M_4(L_\varepsilon) = \frac{3003003000}{1001003001001} > \frac{299}{100000}, \quad (1)$$

$$B_4(L_\varepsilon) < \frac{23}{10000}. \quad (2)$$

Consequently $R_4(L_\varepsilon) > 13/10 > 4/\pi$.

Theorem 1.2 (No universal finite constant). *For every integer $r \geq 1$ and every $\eta > 0$, there is an admissible set E of $n = 3^r$ distinct nonzero complex numbers such that, at $k = (3^r - 1)/2$,*

$$R_k(E) > \left(\frac{2}{\sqrt{3}}\right)^r - \eta.$$

In particular, $\sup_{n,k,E} R_k(E) = \infty$, where the supremum is taken over the assumptions and quantifiers of IE-16.

Theorem 1.1 alone completely resolves IE-16. Theorem 1.2 is stronger; its proof does not rely on numerical experiments. No minimality of nine as a counterexample size, nor an optimal growth rate in n , is asserted.

2 Two elementary interpolation formulas

We will also use the monic approximation problem

$$T_d(K) = \min\{\|q\|_K : q \in \mathbb{C}[z] \text{ is monic of degree } d\}$$

for a finite set $K \subset \mathbb{C}$ with $|K| \geq d + 1$. Here K is allowed to contain zero. Set

$$b_d(K) = \max_{U \subseteq K, |U|=d+1} T_d(U), \quad \Gamma_d(K) = \frac{T_d(K)}{b_d(K)}.$$

These quantities are positive and their minima exist by the same finite-dimensional argument.

Lemma 2.1 (Interpolation identities). *For $S = \{z_0, \dots, z_k\} \subset \mathbb{C} \setminus \{0\}$ with distinct points,*

$$M_k(S) = \left(\sum_{j=0}^k \prod_{h \neq j} \frac{|z_h|}{|z_j - z_h|} \right)^{-1}. \quad (3)$$

For $U \subset \mathbb{C}$ with $|U| = d + 1$, let

$$A_d(U) = \sum_{x \in U} \frac{1}{\prod_{y \in U \setminus \{x\}} |x - y|}.$$

Then

$$T_d(U) = \frac{1}{A_d(U)}. \quad (4)$$

Proof. Write $\ell_j(z) = \prod_{h \neq j} (z - z_h)/(z_j - z_h)$. Interpolation gives $1 = p(0) = \sum_j \ell_j(0)p(z_j)$, so $\|p\|_S \geq 1/\sum_j |\ell_j(0)|$. Equality is attained by interpolating the values

$$p(z_j) = \frac{\overline{\ell_j(0)}}{|\ell_j(0)| \sum_h |\ell_h(0)|}.$$

All $\ell_j(0)$ are nonzero. This proves (3). For the monic problem, replace evaluation at zero by extraction of the leading coefficient. The leading coefficient of the cardinal polynomial at $x \in U$ is $\alpha_x = 1/\prod_{y \neq x} (x - y)$; interpolation of $\overline{\alpha_x}/(|\alpha_x| \sum_y |\alpha_y|)$ has leading coefficient one and norm $1/\sum_y |\alpha_y|$. The matching lower bound follows from the triangle inequality. $\square$

3 The exact nine-point counterexample

For this section let $0 < \varepsilon \leq 1/10$, and retain the definition of L_ε in Theorem 1.1. Its three clusters are

$$C_a = \{\omega^a + \varepsilon\omega^b : 0 \leq b \leq 2\}, \quad a = 0, 1, 2.$$

Points in one cluster have distance $\sqrt{3}\varepsilon$. Points in different clusters have distance at least $\sqrt{3} - 2\varepsilon > 0$. Also $1 - \varepsilon \leq |z| \leq 1 + \varepsilon$, so the nine points are distinct and nonzero.

3.1 The full-set minimum

Proposition 3.1. *For $0 < \varepsilon \leq 1/10$, put*

$$D = 1 + \varepsilon + 3\varepsilon^2 + \varepsilon^3 + \varepsilon^4.$$

Then

$$M_4(L_\varepsilon) = \frac{3\varepsilon(1 + \varepsilon + \varepsilon^2)}{D}, \quad p_*(z) = 1 - \frac{1 + \varepsilon}{D}z^3 \quad (5)$$

is a minimizing polynomial. Its modulus equals $M_4(L_\varepsilon)$ at every point of L_ε .

Proof. The set is invariant under multiplication by ω . Averaging a feasible polynomial over this group does not increase its norm, and leaves a polynomial $1 + cz^3$. Conjugation invariance then allows c to be real, again by averaging. The three distinct values of z^3 on L_ε are

$$W_0 = (1 + \varepsilon)^3 = U, \quad W_1 = V + iW, \quad W_2 = V - iW,$$

where

$$V = 1 - \frac{3}{2}\varepsilon - \frac{3}{2}\varepsilon^2 + \varepsilon^3, \quad W = \frac{3\sqrt{3}}{2}(\varepsilon - \varepsilon^2).$$

Thus the squared minimax problem reduces to minimizing

$$\max\{g_0(c), g_1(c)\}, \quad g_0(c) = (1 + Uc)^2, \quad g_1(c) = (1 + Vc)^2 + W^2c^2.$$

Define

$$c_* = -\frac{1 + \varepsilon}{D}, \quad m = \frac{3\varepsilon(1 + \varepsilon + \varepsilon^2)}{D}, \quad H = 1 - 3\varepsilon + \varepsilon^2 - 3\varepsilon^3 + \varepsilon^4.$$

Direct expansion gives

$$\begin{aligned} 1 + Uc_* &= -m, & g_0(c_*) &= g_1(c_*) = m^2, \\ g'_0(c_*) &= -2Um < 0, \\ \frac{1}{2}g'_1(c_*) &= V + c_*(V^2 + W^2) = \frac{3\varepsilon(1 + \varepsilon)H}{2D} > 0. \end{aligned}$$

For the last inequality, $H \geq 1 - 3\varepsilon - 3\varepsilon^3 \geq 697/1000 > 0$. If $c < c_*$, then $1 + Uc < -m$, so $g_0(c) > m^2$. If $c > c_*$, the convex quadratic g_1 is strictly increasing from c_* , so $g_1(c) > m^2$. The global minimum is therefore m^2 , attained at c_* . All three cubic-image residual moduli are m , proving the final assertion. $\square$

3.2 Every five-point subset

Proposition 3.2. *For $0 < \varepsilon \leq 1/10$ and every five-point subset $S \subset L_\varepsilon$,*

$$M_4(S) \leq U_\varepsilon := \frac{\sqrt{3}\varepsilon(\sqrt{3} + 2\varepsilon)^3}{4(1 - \varepsilon)^4}. \quad (6)$$

Proof. Let $s_a = |S \cap C_a|$. Up to permutation, the only possible profiles are $(2, 2, 1)$, $(3, 1, 1)$ and $(3, 2, 0)$. For any $z_j \in S$, the numerator of its absolute Lagrange coefficient in (3) is at least $(1 - \varepsilon)^4$.

For profile $(2, 2, 1)$, each of the four points in the two two-point clusters has one within-cluster denominator factor $\sqrt{3}\varepsilon$ and three cross-cluster factors at most $\sqrt{3} + 2\varepsilon$. Their four contributions give

$$\frac{1}{M_4(S)} \geq \frac{4(1 - \varepsilon)^4}{\sqrt{3}\varepsilon(\sqrt{3} + 2\varepsilon)^3},$$

which is (6).

For either profile containing a three-point cluster, the three points in that cluster alone give

$$\frac{1}{M_4(S)} \geq \frac{3(1 - \varepsilon)^4}{(\sqrt{3}\varepsilon)^2(\sqrt{3} + 2\varepsilon)^2} = \frac{(1 - \varepsilon)^4}{\varepsilon^2(\sqrt{3} + 2\varepsilon)^2}.$$

Thus

$$M_4(S) \leq \frac{\varepsilon^2(\sqrt{3} + 2\varepsilon)^2}{(1 - \varepsilon)^4} \leq U_\varepsilon,$$

since $4\varepsilon \leq \sqrt{3}(\sqrt{3} + 2\varepsilon)$; indeed $4\varepsilon \leq 2/5 < 3$. These cases exhaust all subsets. $\square$

Proof of Theorem 1.1. Substitute $\varepsilon = 1/1000$ in (5) to obtain the exact rational value (1). It exceeds $299/100000$. For the subset bound, expand

$$\sqrt{3}(\sqrt{3} + 2\varepsilon)^3 = 9 + 18\sqrt{3}\varepsilon + 36\varepsilon^2 + 8\sqrt{3}\varepsilon^3 < 9 + 36\varepsilon + 36\varepsilon^2 + 16\varepsilon^3.$$

At $\varepsilon = 1/1000$, an entirely rational comparison gives

$$\frac{9 + 36/1000 + 36/10^6 + 16/10^9}{4(999/1000)^4} < \frac{23}{10}.$$

Proposition 3.2 now implies (2). Finally,

$$R_4(L_\varepsilon) > \frac{299/100000}{23/10000} = \frac{13}{10} > \frac{4}{\pi},$$

where the last inequality follows already from $\pi > 31/10$. Here $n = 9$ and $1 \leq 4 \leq 7 = n - 2$, so every assumption in IE-16 is met. $\square$

3.3 The limiting ratio is $4/3$

Proposition 3.3. *For the preceding family, $R_4(L_\varepsilon) \rightarrow 4/3$ as $\varepsilon \downarrow 0$.*

Proof. Equation (5) gives $M_4(L_\varepsilon)/\varepsilon \rightarrow 3$. For any fixed subset pattern of type $(2, 2, 1)$, each of the four points with one within-cluster companion has absolute Lagrange coefficient asymptotic to

$$\frac{1}{(\sqrt{3}\varepsilon)(\sqrt{3})^3} = \frac{1}{9\varepsilon}.$$

The remaining coefficient is bounded. Consequently $M_4(S)/\varepsilon \rightarrow 9/4$. For the other two profiles, Proposition 3.2's proof gives $M_4(S) = O(\varepsilon^2)$. There are only 126 fixed subset patterns, so the maximum of their rescaled values tends to $9/4$. Taking the ratio proves the claim. $\square$

4 A three-cluster amplification lemma

The distinction between residual and monic approximation is important in the iteration below. We prove both limits rather than identify the two problems without justification.

Lemma 4.1 (Cluster amplification). *Let $d \geq 1$, and let $K \subset \mathbb{C}$ be a finite set of $m \geq d + 1$ distinct points. Set $a_j = \omega^j$, $j = 0, 1, 2$, and for $\varepsilon > 0$ define*

$$E_\varepsilon = \bigcup_{j=0}^2 (a_j + \varepsilon K), \quad k = 3d + 1.$$

For sufficiently small ε , these are $3m$ distinct nonzero points. As $\varepsilon \downarrow 0$,

$$\frac{T_k(E_\varepsilon)}{\varepsilon^d} \longrightarrow 3^d T_d(K), \quad \frac{M_k(E_\varepsilon)}{\varepsilon^d} \longrightarrow 3^d T_d(K), \quad (7)$$

$$\frac{b_k(E_\varepsilon)}{\varepsilon^d} \longrightarrow 3^d \frac{\sqrt{3}}{2} b_d(K), \quad \frac{B_k(E_\varepsilon)}{\varepsilon^d} \longrightarrow 3^d \frac{\sqrt{3}}{2} b_d(K). \quad (8)$$

In particular,

$$\Gamma_k(E_\varepsilon) \longrightarrow \frac{2}{\sqrt{3}} \Gamma_d(K), \quad R_k(E_\varepsilon) \longrightarrow \frac{2}{\sqrt{3}} \Gamma_d(K). \quad (9)$$

Proof. Let $P(z) = z^3 - 1$ and $t = T_d(K)$. Cluster separation and avoidance of zero follow for sufficiently small ε because K is finite and the three centers are distinct and nonzero. We prove the four limits in three steps.

Step 1: upper bounds for the two full-set minima. Choose a monic minimizing polynomial

$$Q(x) = x^d + \sum_{q=0}^{d-1} q_q x^q, \quad \|Q\|_K = t.$$

For the monic problem take $h(z) = z$; for the residual problem take $h(z) = (-1)^d$. In both cases $|h(a_j)| = 1$. Let $F_0 = P^d h$ and

$$A_j = (P'(a_j))^d h(a_j) = (3a_j^2)^d h(a_j), \quad |A_j| = 3^d.$$

There is a unique polynomial H_ε of degree at most $3d - 1$ with Hermite data

$$\frac{H_\varepsilon^{(q)}(a_j)}{q!} = A_j q_q \varepsilon^{d-q}, \quad 0 \leq q < d, \quad j = 0, 1, 2. \quad (10)$$

Existence and uniqueness follow because the homogeneous system would have a polynomial of degree at most $3d - 1$ divisible by P^d . The inverse interpolation map is fixed, independent of ε . All data in (10) are $O(\varepsilon)$, so every coefficient of H_ε is $O(\varepsilon)$.

Put $F_\varepsilon = F_0 + H_\varepsilon$. Taylor expansion, uniformly for the finite set $x \in K$, gives

$$\varepsilon^{-d} F_\varepsilon(a_j + \varepsilon x) \longrightarrow A_j Q(x). \quad (11)$$

Indeed, the terms of orders below d are prescribed by (10), the order- d coefficient tends to A_j , and all higher terms vanish after rescaling. In the monic case F_ε is monic of degree $3d + 1$. In the residual case $F_\varepsilon(0) = 1 + O(\varepsilon)$, so division by $F_\varepsilon(0)$ gives a feasible residual polynomial without changing the limit. Thus both limsups in (7) are at most $3^d t$.

Step 2: matching full-set lower bounds. Let f_ε be a minimizer of either full-set problem. Step 1 gives $\|f_\varepsilon\|_{E_\varepsilon} = O(\varepsilon^d)$. We first justify boundedness of its coefficients, which is needed for the confluent limit.

Choose distinct $\xi_0, \dots, \xi_d \in K$. At centers a_0, a_1 consider divided differences of orders $q = 0, \dots, d$ on $a_j + \varepsilon\xi_0, \dots, a_j + \varepsilon\xi_q$; at center a_2 use orders $q = 0, \dots, d-1$. There are $3d+2$ functionals, the dimension of polynomials of degree at most $3d+1$. Their matrices on polynomial coefficients converge to the Hermite matrix with multiplicities $(d+1, d+1, d)$. That limiting matrix is invertible: a polynomial in its kernel would have $3d+2$ zeros counted with multiplicity and degree at most $3d+1$. The inverse matrices are consequently uniformly bounded for small ε .

An order- q divided difference of f_ε on these nodes has the form

$$\varepsilon^{-q} \sum_{u=0}^q \frac{f_\varepsilon(a_j + \varepsilon\xi_u)}{\prod_{v \neq u} (\xi_u - \xi_v)},$$

so it is $O(\varepsilon^{d-q})$, hence bounded for $q \leq d$. Uniform boundedness of the inverse matrices proves boundedness of all coefficients of f_ε .

Now expand locally as

$$g_{j,\varepsilon}(x) = \varepsilon^{-d} f_\varepsilon(a_j + \varepsilon x).$$

Terms of degree greater than d tend to zero, because the unscaled coefficients are bounded. The degree-at-most- d truncations have bounded values at $\xi_0, \dots, \xi_d$, so their coefficients are bounded by fixed Vandermonde interpolation. Passing to a subsequence gives both a coefficient limit f_0 and degree-at-most- d limits g_j . For $q < d$ the local bounds imply $f_\varepsilon^{(q)}(a_j) = O(\varepsilon^{d-q}) \rightarrow 0$. Therefore

$$f_0 = P^d h, \quad \deg h \leq 1,$$

and the leading coefficient of g_j is $(3a_j^2)^d h(a_j)$. In the monic case h is monic linear; in the residual case $h(0) = (-1)^d$.

In either case $\max_j |h(a_j)| \geq 1$. For $h(z) = z + c$ this follows from

$$\frac{1}{3} \sum_{j=0}^2 |a_j + c|^2 = 1 + |c|^2;$$

for $h(z) = (-1)^d + bz$ it follows from the analogous identity $1 + |b|^2$. By the definition of the monic minimum,

$$\|g_j\|_K \geq 3^d |h(a_j)| t,$$

including the case of zero leading coefficient, when the right side is zero. Taking the maximum and a subsequence attaining the original liminf gives a lower bound $3^d t$ for both liminfs. This proves (7).

Step 3: all subset limits. Label a $(k+1)$ -point subset by its fixed choices inside the three copies of K , with occupancy (s_0, s_1, s_2) and sum $3d+2$. There are finitely many such labeled patterns. If some $s_j \geq d+2$, a term of the monic Lagrange sum has order $\varepsilon^{-(s_j-1)}$ with a positive limiting coefficient. The associated subset minimum is therefore $o(\varepsilon^d)$. The same holds for residual interpolation, because its additional numerator $\prod_{h \neq j} |z_h|$ tends to one.

If all $s_j \leq d+1$, their sum forces the profile $(d+1, d+1, d)$. Let $U, V \subset K$ be the two $(d+1)$ -point choices. For a point in either full cluster, there are d within-cluster denominator

factors and $2d + 1$ cross-cluster factors. Since every distance between distinct centers is $\sqrt{3}$, the rescaled monic Lagrange sum tends to

$$\frac{A_d(U) + A_d(V)}{(\sqrt{3})^{2d+1}} = \frac{A_d(U) + A_d(V)}{3^d \sqrt{3}}. \quad (12)$$

Contributions from the d -point cluster vanish after multiplication by ε^d . Formula (12) is also the residual limit, again because all numerator moduli tend to one.

The two full-cluster choices are independent. Their smallest possible sum is $2 \min_{|U|=d+1} A_d(U) = 2/b_d(K)$, attained by choosing a minimizing subset in each of those two clusters. Any d points may be used in the third cluster. Taking inverses in (12), and then the maximum over the finitely many patterns, proves (8). Division yields (9). $\square$

5 Iterating the amplification

Proof of Theorem 1.2. Let $K_1 = \{1, \omega, \omega^2\}$ and $d_1 = 1$. Averaging gives $M_1(K_1) = 1$, while $\frac{1}{3} \sum_j |a_j + c|^2 = 1 + |c|^2$ gives $T_1(K_1) = 1$. Every two-point subset has monic and residual minimum $\sqrt{3}/2$, by Lemma 2.1. Hence

$$\Gamma_1(K_1) = R_1(K_1) = c, \quad c = 2/\sqrt{3} > 1.$$

We prove inductively that for every $r \geq 1$ and every $\eta > 0$ there is a set K_r of 3^r distinct nonzero points such that, for $d_r = (3^r - 1)/2$, both

$$\Gamma_{d_r}(K_r) > c^r - \eta, \quad R_{d_r}(K_r) > c^r - \eta.$$

The base case was just established. For the induction step, first choose K_{r-1} with monic ratio greater than $c^{r-1} - \eta/(2c)$. Apply Lemma 4.1 to this fixed set and take the new scale ε sufficiently small that both limiting ratios are approximated within $\eta/2$. Their common limit is $c\Gamma_{d_{r-1}}(K_{r-1}) > c^r - \eta/2$. The new degree is $3d_{r-1} + 1 = d_r$ and the new cardinality is $3 \cdot 3^{r-1} = 3^r$. This proves the induction.

The construction has the recursive form

$$K_r = \bigcup_{a=0}^2 (\omega^a + \varepsilon_r K_{r-1}),$$

where each scale is chosen after the inner set is fixed. No interchange of an infinite-depth limit and a small-scale limit is used. For every fixed r , all scales are positive and the resulting set is finite. The scales may also be chosen rational, since the required strict inequalities hold for every sufficiently small positive scale. Finally $n = 3^r \geq 3$ and $k = (n - 1)/2$ satisfies $1 \leq k \leq n - 2$. Since $c^r \rightarrow \infty$, the residual subset ratios are unbounded. $\square$

Along this sequence of dimensions, the supremum of the ratio is at least n^α , where $\alpha = \log(2/\sqrt{3})/\log 3 > 0$. This is a lower-growth construction, not an assertion that the exponent is optimal. It also shows why merely increasing the proposed numerical constant cannot restore a dimension-independent theorem.

6 An actual normal-GMRES witness and exact verification

6.1 Positive weights certify optimality

The finite counterexample can be checked without relying on a nonlinear optimizer or on an equality between two separately defined GMRES bounds. Keep D from Proposition 3.1, and set

$$H = 1 - 3\varepsilon + \varepsilon^2 - 3\varepsilon^3 + \varepsilon^4, \quad Q = (1 + \varepsilon)^2(1 + \varepsilon + \varepsilon^2),$$

$$\mu_0 = \frac{H}{3D}, \quad \mu_1 = \mu_2 = \frac{Q}{3D}, \quad \nu_{a,b} = \frac{1}{3}\mu_{(b-a) \bmod 3}.$$

For $0 < \varepsilon \leq 1/10$, these weights are positive and sum to one; the latter follows from $H + 2Q = 3D$. Writing $z_{a,b} = \omega^a + \varepsilon\omega^b$, direct algebra gives

$$\sum_{a,b} \nu_{a,b} \overline{p_*(z_{a,b})} z_{a,b}^\ell = 0, \quad \ell = 1, 2, 3, 4. \quad (13)$$

For $\ell \not\equiv 0 \pmod{3}$ the sum vanishes by rotation. For $\ell = 3$, conjugation cancels the imaginary parts, and the identity is the cancellation of

$$\mu_0 U(1 + c_* U) + (1 - \mu_0)(V + c_*(V^2 + W^2)) = 0$$

using the expressions in Proposition 3.1.

Let $m = M_4(L_\varepsilon)$. For any feasible p , the difference $p - p_*$ has no constant term, so (13) implies the exact identity

$$\sum_{a,b} \nu_{a,b} |p(z_{a,b})|^2 = m^2 + \sum_{a,b} \nu_{a,b} |p(z_{a,b}) - p_*(z_{a,b})|^2 \geq m^2. \quad (14)$$

Together with $|p_*(z_{a,b})| = m$, this is a second proof of the full-set value.

Now take the nonsingular normal matrix

$$A = \text{diag}(z_{0,0}, z_{0,1}, z_{0,2}, z_{1,0}, \dots, z_{2,2})$$

and the unit initial residual with entries $(r_0)_{a,b} = \sqrt{\nu_{a,b}}$. The defining residual minimization for GMRES at step four is exactly the square root of the left side of (14), minimized over feasible p . Its value is m . Conversely, p_* gives a residual of norm m for *every* unit initial residual, since its nine diagonal residual entries all have modulus m . Thus m is indeed the worst-case step-four norm for this explicit normal matrix.

6.2 What the computational certificate checks

The program `code/verify.py` requires only Python's standard library. For $\varepsilon = 1/1000$, it represents a node as $a + b\omega$ with rational a, b and uses

$$|a + b\omega|^2 = a^2 - ab + b^2.$$

It checks positivity and normalization of the nine weights, all nine residual moduli squared, and each of the four complex identities (13) exactly. For each of the 126 subsets it forms each squared absolute Lagrange weight as a rational number and encloses its square root by integer square roots. Rational alternating-series bounds in Machin's identity enclose π . Every strict inequality is compared as a rational inequality, without a floating-point tolerance.

The generated certificate gives

$$\begin{aligned} M_4(L_{1/1000}) &= 0.002999994003011984982048\dots, \\ B_4(L_{1/1000}) &= 0.002250559965550380002031\dots, \\ R_4(L_{1/1000}) &= 1.332998919794758294237942\dots, \\ M_4(L_{1/1000}) - \frac{4}{\pi}B_4(L_{1/1000}) &> 0.0001344920570754357296. \end{aligned}$$

The first three decimal displays are descriptive; the exact rational interval endpoints are stored in `certificates/all_subset_certificate.json`. The profile counts are 81 for $(2, 2, 1)$, 27 for $(3, 1, 1)$ and 18 for $(3, 2, 0)$. The analytic proof above does not depend on this enumeration. The verification program certifies the finite witness; it is not a formal proof-assistant verification of the general amplification theorem.

Sources

- [1] Alex Townsend, *Open Problems in Numerical Linear Algebra*, IE-16, “A sharp subset bound for worst-case normal GMRES.” Canonical statement accessed 12 September 2026. The page inspected lists “Partially resolved” and a literature-check date of 10 September 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/linear-systems-and-elimination/IE-16/README.md>
- [2] Jörg Liesen and Petr Tichý, *The worst-case GMRES for normal matrices*, BIT Numerical Mathematics **44** (2004), 79–98. The subset formula is (3.10); the original conjecture and candidate constant are in §3.2.2, (3.16), pp. 91–92. Author-hosted full text: <https://page.math.tu-berlin.de/~liesen/Publicat/LieTic04.pdf>